

Labwork 2: Point processing, Image Filtering and Noise Removal

Part A

Task A1:

$$I = \begin{bmatrix} 104 & 100 & 108 \\ 99 & 106 & 98 \\ 95 & 90 & 85 \end{bmatrix}$$

- Applying $K_1 = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ to I , we'll get

$$O_1 = \begin{bmatrix} 98.3 & 98.3 & 98.3 \\ 98.3 & 98.3 & 98.3 \\ 98.3 & 98.3 & 98.3 \end{bmatrix} \Rightarrow O_1 = 98.333$$

- Applying $K_2 = \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$ to I , we'll get

$$O_2 = \frac{1}{16} (104 + 2 \cdot 100 + 108 + 2 \cdot 99 + 4 \cdot 106 + 2 \cdot 98 + 95 + 2 \cdot 90 + 85) = 99.375$$

$O_1 < O_2 \Rightarrow K_2$ gives more importance to the center pixel because the center pixel has the most weight = 4

Task A2:

$$I = \begin{bmatrix} 12 & 13 & 14 \\ 15 & 255 & 16 \\ 14 & 13 & 12 \end{bmatrix}$$

Applying 3×3 median filter

$$\Rightarrow [12 \ 12 \ 13 \ 13 \ 14 \ 14 \ 15 \ 255 \ 16]$$

new center pixel



- Median filter is effective for salt-and-pepper noise because the noise doesn't affect the filter that much when it has a high/low value compare to the rest of the pixels in the matrix, the filter would just pick the middle value from the sorted list

Task A3:

$$I = \begin{bmatrix} 10 & 10 & 10 \\ 10 & 50 & 10 \\ 10 & 10 & 10 \end{bmatrix} \quad K = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

$$\Rightarrow O = -10 - 10 - 10 - 10 + 400 - 10 - 10 - 10 - 10 \\ = 320$$

- This mask respond strongly to isolated bright points because because the mask try to highlight the center pixel and when the center pixel has a high value (bright), it will be higher (brighter)

Task A4:

$$J = \begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 100 & 100 & 100 \end{bmatrix} \quad G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

$$G_x = -10 + 10 - 40 + 40 - 100 + 100 = 0$$

$$G_y = -10 - 20 - 10 + 100 + 200 + 100 = 360$$

$$G = \sqrt{G_x^2 + G_y^2} = 360$$

Based on the values of G_x and G_y , the main edge in J is the ~~bottom~~ ^{vertical} edge horizontal edge because G_x places more weights on the vertical edge and its output is 0 while G_y places



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more weights on the horizontal edge and its out is $360 > 0$

Task A5:

$$J = \begin{bmatrix} 10 & 10 & 10 \\ 10 & 50 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

$$K = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Applying K to J :

$$O_k = 10 + 10 - 200 + 10 + 10 = -160$$

$$g(1, 1) = 50 + 160 = 210$$